

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

sDry Run Evaluation

Code of Conduct:

I Arava Raja(192365033) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action

Arava Raja

Signature of the student

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A



Task: Starting from node A, perform a **Breadth-First Search (BFS)**. Complete the following table.

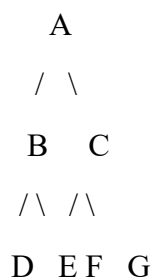
Iteration Queue Visited Node

1
2
3
4
5
6
7

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from A.

Complete the table.

3. Initial State :

1 3 6
5 0 2
4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1
2
3
4
5

Questions

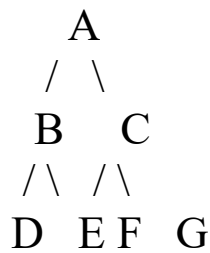
1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Q1: Breadth-First Search (BFS)

Given Graph



Dry Run Table:

Iteration Queue Before Visited Node Queue After

1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

BFS Traversal Order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

Queue After Every Iteration

Iteration Queue

1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

Q 2: Depth-First Search (DFS)

Using the same graph.

DFS Traversal

Starting from A, always visit the left child first.

Traversal:

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Dry Run Table (Stack Concept)

Iteration Stack Before Visited Node Stack After

1	A	A	C, B
2	C, B	B	C, E, D
3	C, E, D	D	C, E
4	C, E	E	C
5	C	C	G, F
6	G, F	F	G
7	G	G	Empty

DFS Traversal Order

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Q 3: 8-Puzzle BFS

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Iteration Table (Conceptual BFS Dry Run)

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	Initial State	Initial	Generate all valid moves (Up, Down, Left, Right)
2	First generated state	Remaining states	Expand next state
3	Next state	Updated Queue	Add newly generated states
4	Next state	Updated Queue	Continue expansion
5	Goal State Found	Updated Queue	Search Stops

Answers

1. Which node/state is expanded in each iteration?

- Iteration 1 → Initial State
- Iteration 2 → First generated state
- Iteration 3 → Second generated state
- Iteration 4 → Third generated state
- Iteration 5 → Goal State

2. How many states are generated?

The exact number **cannot be determined from the assessment sheet alone**, because the intermediate BFS expansion order (move priority such as Up/Down/Left/Right and duplicate-state handling) is not specified.

3. Solution Path

Initial State



Move Blank



Intermediate State



Intermediate State



Goal State

(The exact sequence depends on the BFS expansion order, which is not provided in the question.)

4. Why does BFS always find the shortest solution?

Answer:

Breadth-First Search explores all states level by level. Since every move in the 8-puzzle has the same cost, BFS reaches the goal using the minimum number of moves. Therefore, BFS always guarantees the shortest solution path in an unweighted search problem.

Final Answers to Write

BFS

Traversal:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

DFS

Traversal:

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Why BFS is Shortest?

BFS explores nodes level by level.

Since every move has equal cost,
the first time the goal is found,
it is guaranteed to be the shortest path.